

Development of Wireless Master-Slave Humanoid Robot Arm

Authors: Nourhan A. Abbass, Yousri M. Diab, Eslam H. Mersal, Ahmed M. Hanafi, Abdelrady O. Elnady (2023)

ABSTRACT

This paper describes the development of a wireless master-slave control system for a humanoid robotic arm. The system reads joint configurations from a master glove worn by an operator and transmits coordinates wirelessly to a slave robotic arm. The control loop is optimized to minimize latency and ensure high accuracy in replication.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Supports the design of wireless interfaces and real-time synchronization between separate control nodes in master-slave architectures.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Nourhan A. Abbass provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.